

▼ Aerofit case study

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

```
aerofit_data = pd.read_csv('aerofit_treadmill.csv')
```

```
aerofit_data.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	
0	KP281	18	Male	14	Single	3	4	29562	112	
1	KP281	19	Male	15	Single	2	3	31836	75	
2	KP281	19	Female	14	Partnered	4	3	30699	66	
3	KP281	19	Male	12	Single	3	3	32973	85	
4	KP281	20	Male	13	Partnered	4	2	35247	47	

Next steps:

[Generate code with aerofit_data](#)
[New interactive sheet](#)

```
aerofit_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product         180 non-null   object
1   Age             180 non-null   int64
2   Gender          180 non-null   object
3   Education       180 non-null   int64
4   MaritalStatus  180 non-null   object
5   Usage           180 non-null   int64
6   Fitness         180 non-null   int64
7   Income          180 non-null   int64
8   Miles           180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
aerofit_data.describe()
```

	Age	Education	Usage	Fitness	Income	Miles	
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000	
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444	
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605	
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000	
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000	
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000	
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000	
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000	

```
cat_cols = ['Product Purchased', 'Gender', 'MaritalStatus']
for c in cat_cols:
    if c in aerofit_data.columns:
        aerofit_data[c] = aerofit_data[c].astype('category')

print(cat_cols)
```

```
['Product Purchased', 'Gender', 'MaritalStatus']
```

```
aerofit_data.describe(include=['category', 'object'])
```



	Product	Gender	MaritalStatus	
count	180	180	180	
unique	3	2	2	
top	KP281	Male	Partnered	
freq	80	104	107	

```
for col in aerofit_data.columns:
    print(f"\nValue counts for {col}:")
    print(aerofit_data[col].value_counts(dropna=False))
```

```
1      2
Name: count, dtype: int64

Value counts for Income:
Income
45480    14
52302     9
53439     8
54576     8
46617     8
..
85906     1
99601     1
103336    1
95866     1
95508     1
Name: count, Length: 62, dtype: int64
```

Value counts for Miles:

```
Miles
85      27
95      12
66      10
75      10
47       9
106      9
113       8
94        8
53        7
100       7
200       6
56        6
64        6
180       6
160       5
127       5
42        4
150       4
74        3
38        3
103       3
170       3
120       3
132       2
141       2
112       1
212       1
188       1
169       1
80        1
140       1
21        1
240       1
300       1
280       1
260       1
360       1
Name: count, dtype: int64
```

▼ Finding the missing value

```
aerofit_data.isnull().sum()
```

	0
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0

```
num_cols = aerofit_data.select_dtypes(include=[np.number]).columns.tolist()
print('\nNumeric columns:', num_cols)
```

```
Income      0
Numeric columns: ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']
Miles      0
```

✓ Z-score method

```
z_outliers = {}
for c in num_cols:
    z = np.abs(stats.zscore(aerofit_data[c].dropna()))
    z_outliers[c] = (z > 3).sum()
print('\nOutliers (z-score > 3 count per numeric column):')
print(z_outliers)
```

```
Outliers (z-score > 3 count per numeric column):
{'Age': np.int64(1), 'Education': np.int64(3), 'Usage': np.int64(2), 'Fitness': np.int64(0), 'Income': np.int64(3), 'Miles': np.int64(0)}
```

✓ Visual analysis

```
for c in num_cols:
    plt.figure()
    sns.histplot(aerofit_data[c].dropna(), kde=True)
    plt.title(f'Histogram + KDE: {c}')
    plt.savefig(f'fig_{c}_hist.png')
    plt.close()
```

```
plt.figure()
sns.boxplot(x=aerofit_data[c].dropna())
plt.title(f'Boxplot: {c}')
plt.savefig(f'fig_{c}_box.png')
plt.close()
```

New Section